

Type- 4 : interval $(-L, L)$ **1.** Find the Fourier expansion of $f(x) = e^{ax}$ in $(l, -l)$ **Solution:** Let $f(x) = \frac{a_0}{2} + \sum a_n \cos \frac{n\pi x}{l} + \sum b_n \sin \frac{n\pi x}{l}$ (1)

$$\therefore a_0 = \frac{1}{l} \int_{-l}^l f(x) dx = \frac{1}{l} \int_{-l}^l e^{ax} dx = \frac{1}{l} \left[\frac{e^{ax}}{a} \right]_{-l}^l = \frac{1}{al} [e^{al} - e^{-al}] = \frac{1}{al} \left[\frac{e^{al} - e^{-al}}{2} \right] = \frac{2 \sinh al}{al}$$

$$\begin{aligned} \therefore a_n &= \frac{1}{l} \int_{-l}^l f(x) \cos \frac{n\pi x}{l} dx = \frac{1}{l} \int_{-l}^l e^{ax} \cos \frac{n\pi x}{l} dx \\ &= \frac{1}{l} \left[\frac{1}{a^2 + n^2 \pi^2 / l^2} \left\{ e^{ax} \left(a \cos \frac{n\pi x}{l} + \frac{n\pi}{l} \sin \frac{n\pi x}{l} \right) \right\} \right]_{-l}^l \\ &= \frac{l}{a^2 l^2 + n^2 \pi^2} [e^{al}(a \cos n\pi + 0) - e^{-al}(a \cos n\pi + 0)] \\ &= \frac{al}{a^2 l^2 + n^2 \pi^2} [e^{al}(-1)^n - e^{-al}(-1)^n] \\ &= \frac{2al(-1)^n}{a^2 l^2 + n^2 \pi^2} \left[\frac{e^{al} - e^{-al}}{2} \right] = \frac{2al(-1)^n}{a^2 l^2 + n^2 \pi^2} \sinh al \end{aligned}$$

$$\begin{aligned} \therefore b_n &= \frac{1}{l} \int_{-l}^l f(x) \sin \frac{n\pi x}{l} dx = \frac{1}{l} \int_{-l}^l e^{ax} \sin \frac{n\pi x}{l} dx \\ &= \frac{1}{l} \left[\frac{1}{a^2 + n^2 \pi^2 / l^2} \left\{ e^{ax} \left(a \sin \frac{n\pi x}{l} - \frac{n\pi}{l} \cos \frac{n\pi x}{l} \right) \right\} \right]_{-l}^l \\ &= \frac{l}{a^2 l^2 + n^2 \pi^2} [e^{al} \left(0 - \frac{n\pi}{l} \cos n\pi \right) - e^{-al} \left(0 - \frac{n\pi}{l} \cos n\pi \right)] \\ &= \frac{l}{a^2 l^2 + n^2 \pi^2} [e^{al} \left(0 - \frac{n\pi}{l} (-1)^n \right) - e^{-al} \left(0 - \frac{n\pi}{l} (-1)^n \right)] \\ &= \frac{l}{a^2 l^2 + n^2 \pi^2} \cdot \frac{n\pi}{l} \cdot (-1)^{n+1} (e^{al} - e^{-al}) \\ &= \frac{2n\pi(-1)^{n+1}}{a^2 l^2 + n^2 \pi^2} \cdot \left(\frac{e^{al} - e^{-al}}{2} \right) = \frac{2n\pi(-1)^{n+1}}{a^2 l^2 + n^2 \pi^2} \cdot \sinh al \end{aligned}$$

Putting these values in (1)

$$\begin{aligned} f(x) &= \frac{\sinh al}{al} + \sum \frac{2al(-1)^n}{a^2 l^2 + n^2 \pi^2} \sinh al \cos \frac{n\pi x}{l} + \sum \frac{2n\pi(-1)^{n+1}}{a^2 l^2 + n^2 \pi^2} \cdot \sinh al \sin \frac{n\pi x}{l} \\ &= \frac{\sinh al}{al} + 2al \sinh al \sum \frac{(-1)^n}{a^2 l^2 + n^2 \pi^2} \cos \frac{n\pi x}{l} + 2\pi \sinh al \sum \frac{n(-1)^{n+1}}{a^2 l^2 + n^2 \pi^2} \sin \frac{n\pi x}{l} \end{aligned}$$

2. Find the Fourier expansion of $f(x) = \begin{cases} 2, & -2 < x < 0 \\ x, & 0 < x < 2 \end{cases}$ **Solution:** Comparing the interval $(-2, 2)$ with $(-l, l)$ we find that here $l = 2$

$$\begin{aligned} \text{Let } f(x) &= \frac{a_0}{2} + \sum a_n \cos \frac{n\pi x}{l} + \sum b_n \sin \frac{n\pi x}{l} \\ &= \frac{a_0}{2} + \sum a_n \cos \frac{n\pi x}{2} + \sum b_n \sin \frac{n\pi x}{2} \end{aligned} \quad \text{..... (1)}$$

$$\begin{aligned} \therefore a_0 &= \frac{1}{l} \int_{-l}^l f(x) dx = \frac{1}{2} \int_{-2}^2 f(x) dx = \frac{1}{2} \left[\int_{-2}^0 2 dx + \int_0^2 x dx \right] = \frac{1}{2} \left[\{2x\}_{-2}^0 + \left\{ \frac{x^2}{2} \right\}_0^2 \right] \\ &= \frac{1}{2} [(0 + 4) + (2 - 0)] = \frac{6}{2} = 3 \end{aligned}$$

$$\begin{aligned} a_n &= \frac{1}{l} \int_{-l}^l f(x) \cos \frac{n\pi x}{l} dx = \frac{1}{2} \int_{-2}^2 f(x) \cos \frac{n\pi x}{2} dx \\ &= \frac{1}{2} \left[\int_{-2}^0 2 \cos \frac{n\pi x}{2} dx + \int_0^2 x \cos \frac{n\pi x}{2} dx \right] \\ &= \frac{1}{2} \left[\left\{ 2 \cdot \frac{2}{n\pi} \sin \frac{n\pi x}{2} \right\}_{-2}^0 + \left\{ (x) \cdot \left(\frac{2}{n\pi} \sin \frac{n\pi x}{2} \right) - (1) \left(-\frac{4}{n^2 \pi^2} \cos \frac{n\pi x}{2} \right) \right\}_0^2 \right] \end{aligned}$$

$$= \frac{1}{2} \left[\{0 - 0\} + \left\{ 0 + \frac{4}{n^2 \pi^2} \cos n\pi - \frac{4}{n^2 \pi^2} \right\} \right]$$

$$= \frac{2}{n^2 \pi^2} \{ \cos n\pi - 1 \} = \begin{cases} 0 & \text{when } n \text{ is even} \\ -\frac{4}{n^2 \pi^2} & \text{when } n \text{ is odd} \end{cases}$$

$$b_n = \frac{1}{l} \int_{-l}^l f(x) \sin \frac{n\pi x}{l} dx = \frac{1}{2} \int_{-2}^2 f(x) \sin \frac{n\pi x}{2} dx$$

$$= \frac{1}{2} \left[\int_{-2}^0 2 \sin \frac{n\pi x}{2} dx + \int_0^2 x \sin \frac{n\pi x}{2} dx \right]$$

$$= \frac{1}{2} \left[\left\{ 2 \cdot \left(-\frac{2}{n\pi} \cos \frac{n\pi x}{2} \right) \right\}_{-2}^0 + \left\{ (x) \cdot \left(-\frac{2}{n\pi} \cos \frac{n\pi x}{2} \right) - (1) \left(-\frac{4}{n^2 \pi^2} \sin \frac{n\pi x}{2} \right) \right\}_0^2 \right]$$

$$= \frac{1}{2} \left[\left\{ -\frac{4}{n\pi} + \frac{4}{n\pi} \cos n\pi \right\} + \left\{ (2) \left(-\frac{2}{n\pi} \cos n\pi \right) + 0 \right\} \right]$$

$$= \frac{1}{2} \left\{ -\frac{4}{n\pi} + \frac{4}{n\pi} \cos n\pi - \frac{4}{n\pi} \cos n\pi \right\} = -\frac{2}{n\pi}$$

Putting these values in (1)

$$f(x) = \frac{3}{2} - \frac{4}{\pi^2} \left[\frac{1}{1^2} \cos \frac{\pi x}{2} + \frac{1}{3^2} \cos \frac{3\pi x}{2} + \frac{1}{5^2} \cos \frac{5\pi x}{2} + \dots \right] - \frac{2}{\pi} \left[\frac{1}{1} \sin \frac{\pi x}{2} + \frac{1}{2} \sin \frac{2\pi x}{2} + \frac{1}{3} \sin \frac{3\pi x}{2} + \dots \right]$$

3. Expand $f(x) = \begin{cases} 0, & -c < x < 0 \\ a, & 0 < x < c \end{cases}$ in Fourier series of period $2a$

Solution: Let $f(x) = \frac{a_0}{2} + \sum a_n \cos \frac{n\pi x}{l} + \sum b_n \sin \frac{n\pi x}{l}$ and $l = c$ (1)

$$a_0 = \frac{1}{l} \int_{-l}^l f(x) dx = \frac{1}{c} \int_{-c}^c f(x) dx = \frac{1}{c} \left[\int_{-c}^0 0 \cdot dx + \int_0^c a dx \right] = \frac{1}{c} [0 + [ax]_0^c] = \frac{1}{c} \cdot ac = a$$

$$a_n = \frac{1}{l} \int_{-l}^l f(x) \cos \frac{n\pi x}{l} dx = \frac{1}{c} \left[\int_{-c}^0 0 \cdot \cos \frac{n\pi x}{c} dx + \int_0^c a \cos \frac{n\pi x}{c} dx \right] = \frac{1}{c} \left[a \left(\frac{c}{n\pi} \sin \frac{n\pi x}{c} \right) \right]_0^c$$

$$= \frac{a}{n\pi} [0 - 0] = 0$$

$$b_n = \frac{1}{l} \int_{-l}^l f(x) \sin \frac{n\pi x}{l} dx = \frac{1}{c} \left[\int_{-c}^0 0 \cdot \sin \frac{n\pi x}{c} dx + \int_0^c a \sin \frac{n\pi x}{c} dx \right] = \frac{1}{c} \left[a \left(-\frac{c}{n\pi} \right) \cos \frac{n\pi x}{c} \right]_0^c$$

$$= -\frac{a}{n\pi} \{ \cos n\pi - 1 \} = \begin{cases} 0 & \text{if } n \text{ is even} \\ \frac{2a}{n\pi} & \text{if } n \text{ is odd} \end{cases}$$

Putting these values in (1)

$$f(x) = \frac{a}{2} + \frac{2a}{\pi} \left[\frac{1}{1} \sin \frac{\pi x}{c} + \frac{1}{3} \sin \frac{3\pi x}{c} + \frac{1}{5} \sin \frac{5\pi x}{c} + \dots \right]$$

4. A sinusoidal voltage $E \sin \omega t$ is passed through a half – wave rectifier, which clips the negative portion of the wave. Develop the resulting periodic function in a Fourier series, where

$$U(t) = \begin{cases} 0 & \text{when } -\frac{T}{2} < t < 0 \\ E \sin \omega t & \text{when } 0 < t < \frac{T}{2} \end{cases} \text{ and } T = 2\pi/\omega$$

Solution: The period is given by $T = \frac{2\pi}{\omega}$, $\therefore c = \frac{\pi}{\omega}$

Let $u(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \cos \left(\frac{n\pi t}{\pi/\omega} \right) + b_n \sin \left(\frac{n\pi t}{\pi/\omega} \right) \right]$ be the Fourier series

i.e., $u(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} [a_n \cos(n\omega t) + b_n \sin(n\omega t)]$ is the required Fourier series

$$\begin{aligned} a_0 &= \frac{1}{\pi/\omega} \int_{-T/2}^{T/2} u(t) dt = \frac{\omega}{\pi} \left\{ \int_{-T/2}^0 0 dt + \int_0^{T/2} E \sin \omega t dt \right\} \\ &= \frac{\omega E}{\pi} \int_0^{\pi/\omega} \sin \omega t dt = \frac{\omega E}{\pi} \left[-\frac{\cos \omega t}{\omega} \right]_0^{\pi/\omega} \\ &= \frac{-E}{\pi} [\cos \pi - \cos 0] = \frac{2E}{\pi} \\ \therefore a_0 &= \frac{2E}{\pi} \end{aligned}$$

$$\begin{aligned} a_n &= \frac{\omega}{\pi} \int_0^{\pi/\omega} E \sin \omega t \cdot \cos n\omega t dt = \frac{\omega E}{2\pi} \int_0^{\pi/\omega} 2 \sin \omega t \cdot \cos n\omega t dt \\ &= \frac{\omega E}{2\pi} \int_0^{\pi/\omega} [\sin(1+n)\omega t + \sin(1-n)\omega t] dt \\ &= \frac{\omega E}{2\pi} \left[\frac{-\cos(1+n)\omega t}{(1+n)\omega} - \frac{\cos(1-n)\omega t}{(1-n)\omega} \right]_0^{\pi/\omega} \\ &= \frac{\omega E}{2\pi} \left(\frac{-1}{\omega} \right) \left[\frac{\cos(n+1)\omega t}{n+1} - \frac{\cos(n-1)\omega t}{n-1} \right]_0^{\pi/\omega} \\ &= \frac{-E}{2\pi} \left\{ \left[\frac{\cos(n+1)\pi}{n+1} - \frac{\cos(n-1)\pi}{n-1} \right] - \left[\frac{1}{n+1} - \frac{1}{n-1} \right] \right\} \\ &= \frac{-E}{2\pi} \left\{ \left[\frac{-(-1)^n}{n+1} + \frac{(-1)^n}{n-1} \right] - \left[\frac{1}{n+1} - \frac{1}{n-1} \right] \right\} \\ &= \frac{-E}{2\pi} \left\{ \left[\frac{(-1)^{n+1}}{n+1} \right] - \left[\frac{-2}{n^2-1} \right] \right\} \\ &= \frac{-2E}{2\pi} \left[\frac{(-1)^{n+1}}{n^2-1} \right] = \frac{-E}{\pi(n^2-1)} [(-1)^n + 1] \\ \therefore a_n &= \frac{-2E}{\pi(n^2-1)} \quad \text{when } n \text{ is even} \\ &= 0 \quad \text{when } n \text{ is odd, } n \neq 1 \end{aligned}$$

When $n = 1$, we have

$$\begin{aligned} a_1 &= \frac{\omega E}{\pi} \int_0^{\pi/\omega} \sin \omega t \cos \omega t dt = \frac{\omega E}{\pi} \int_0^{\pi/\omega} \sin 2\omega t dt = \frac{\omega E}{\pi} \left[\frac{-\cos 2\omega t}{2\omega} \right]_0^{\pi/\omega} = \frac{-E}{4\pi} [\cos 2\pi - \cos 0] = 0 \\ \therefore a_n &= 0 \text{ for all odd values of } n \\ \therefore a_n &= \frac{-2E}{\pi(4n^2-1)} \quad \{\text{For } n \text{ even, } n \text{ is replaced by } 2n\} \end{aligned}$$

$$\begin{aligned} b_n &= \frac{\omega}{\pi} \int_0^{\pi/\omega} E \sin \omega t \cdot \sin n\omega t dt = \frac{\omega E}{2\pi} \int_0^{\pi/\omega} 2 \sin \omega t \cdot \sin n\omega t dt \\ &= \frac{\omega E}{2\pi} \int_0^{\pi/\omega} [\cos(1-n)\omega t - \cos(1+n)\omega t] dt \\ &= \frac{\omega E}{2\pi} \left[\frac{\sin(1-n)\omega t}{1-n} - \frac{\sin(1+n)\omega t}{1+n} \right]_0^{\pi/\omega} \\ &= \frac{\omega E}{2\pi} \left[\frac{\sin(1-n)\pi}{1-n} - \frac{\sin(1+n)\pi}{1+n} - 0 \right] \\ &= 0 \quad n \neq 1 \end{aligned}$$

When $n = 1$, we have

$$\begin{aligned} b_1 &= \frac{\omega}{\pi} \int_0^{\pi/\omega} E \sin^2 \omega t dt = \frac{\omega E}{2\pi} \int_0^{\pi/\omega} (1 - \cos 2\omega t) dt = \frac{\omega E}{2\pi} \left[t - \frac{\sin 2\omega t}{2\omega} \right]_0^{\pi/\omega} = \frac{\omega E}{2\pi} \left[\frac{\pi}{\omega} - 0 \right] = \frac{E}{2} \\ \therefore b_n &= 0 \text{ when } n \neq 1 \text{ and } b_1 = \frac{E}{2} \end{aligned}$$

Hence the required Fourier series is

$$u(t) = \frac{a_0}{2} + b_1 \sin \omega t + \sum_{n=1}^{\infty} a_n \cos n\omega t$$

$$\text{i.e., } u(t) = \frac{E}{\pi} + \frac{E}{2} \sin \omega t - \frac{2E}{\pi} \sum_{n=1}^{\infty} \frac{2 \cos n\omega t}{4n^2 - 1}$$

or it can be written in the expanded form as

$$u(t) = \frac{E}{\pi} + \frac{E}{2} \sin \omega t - \frac{2E}{\pi} \left[\frac{\cos 2\omega t}{3} + \frac{\cos 4\omega t}{15} + \frac{\cos 6\omega t}{35} + \dots \right]$$

$$\text{i.e., } u(t) = \frac{E}{\pi} + \frac{E}{2} \sin \omega t - \frac{2E}{\pi} \left[\frac{\cos 2\omega t}{1 \times 3} + \frac{\cos 4\omega t}{3 \times 5} + \frac{\cos 6\omega t}{5 \times 7} + \dots \right]$$

5. Obtain the Fourier series for $f(x) = x^2$ from $x = -l$ to $x = l$ and deduce that

$$(i) \quad \frac{\pi^2}{12} = \frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots \quad (ii) \quad \frac{\pi^2}{6} = \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \dots$$

Solution: Since, $f(x) = x^2$ is an even function, we have $b_n = 0$

$$\therefore f(x) = \frac{a_0}{2} + \sum a_n \cos \frac{n\pi x}{l}$$

$$\text{Now, } a_0 = \frac{2}{l} \int_0^l f(x) dx = \frac{2}{l} \int_0^l x^2 dx = \frac{2}{l} \left[\frac{x^3}{3} \right]_0^l = \frac{2l^2}{3}$$

$$\begin{aligned} a_n &= \frac{2}{l} \int_0^l f(x) \cos \frac{n\pi x}{l} dx = \frac{2}{l} \int_0^l x^2 \cos \frac{n\pi x}{l} dx \\ &= \frac{2}{l} \left[x^2 \left(\frac{l}{n\pi} \sin \frac{n\pi x}{l} \right) - (2x) \left(-\frac{l^2}{n^2\pi^2} \cos \frac{n\pi x}{l} \right) + (2) \left(\frac{-l^3}{n^3\pi^3} \sin \frac{n\pi x}{l} \right) \right]_0^l \\ &= \frac{2}{l} \left[\frac{2l^3}{n^2\pi^2} (-1)^n \right] = (-1)^n \frac{4l^2}{n^2\pi^2} \end{aligned}$$

$$\therefore x^2 = \frac{l^2}{3} + \sum (-1)^n \frac{4l^2}{n^2\pi^2} \cos \frac{n\pi x}{l} = \frac{l^2}{3} - \frac{4l^2}{\pi^2} \left[\frac{1}{1^2} \cos \frac{\pi x}{l} - \frac{1}{2^2} \cos \frac{2\pi x}{l} + \frac{1}{3^2} \cos \frac{3\pi x}{l} + \dots \right]$$

(i) Now, put $x = 0$

$$\therefore \frac{\pi^2}{12} = \left[\frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots \right]$$

$$\therefore \frac{\pi^2}{12} = \frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots$$

(ii) Now, put $x = l$, we get,

$$\frac{2l^2}{3} = -\frac{4l^2}{\pi^2} \left[-\frac{1}{1^2} - \frac{1}{2^2} - \frac{1}{3^2} - \frac{1}{4^2} - \dots \right]$$

$$\therefore \frac{\pi^2}{6} = \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots$$

6. Find the Fourier series for $f(x) = 1 - x^2$ when $-1 \leq x \leq 1$.

Solution: We have $f(-x) = 1 - (-x)^2 = 1 - x^2 = f(x)$.

Hence, $f(x)$ is even function $\therefore b_n = 0$

$$\therefore f(x) = \frac{a_0}{2} + \sum a_n \cos \frac{n\pi x}{l}$$

Comparing $(-l, l)$ with $(-1, 1)$ we get $l = 1$

$$\text{Now, } a_0 = \frac{2}{l} \int_0^l f(x) dx = 2 \int_0^1 (1 - x^2) dx = 2 \left[x - \frac{x^3}{3} \right]_0^1 = 2 \left[1 - \frac{1}{3} \right] = \frac{4}{3}$$

$$a_0 = \frac{2}{l} \int_0^l f(x) \cos \frac{n\pi x}{l} dx = 2 \int_0^1 (1 - x^2) \cos n\pi x dx$$

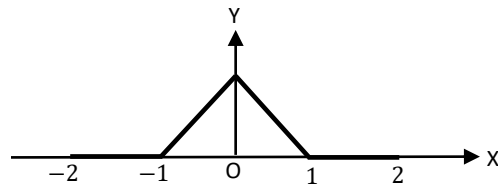
$$\begin{aligned}
&= 2 \left[(1-x^2) \cdot \left(\frac{\sin n\pi x}{n\pi} \right) - (-2x) \left(-\frac{\cos n\pi x}{n^2\pi^2} \right) + (-2) \left(-\frac{\sin n\pi x}{n^3\pi^3} \right) \right]_0^1 \\
&= 2 \left[-\frac{2\cos n\pi}{n^2\pi^2} \right] = \frac{-4(-1)^n}{n^2\pi^2} \\
\text{Hence, } f(x) &= \frac{2}{3} - \frac{4}{\pi^2} \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \cos n\pi x
\end{aligned}$$

7. Obtain the Fourier expansion for $f(x) = \begin{cases} 0, & -2 < x < -1 \\ 1+x, & -1 < x < 0 \\ 1-x, & 0 < x < 1 \\ 0, & 1 < x < 2 \end{cases}$ in the range $(-2, 2)$

Solution: $f(-x) = \begin{cases} 0, & -2 < -x < -1 \\ 1-x, & -1 < -x < 0 \\ 1+x, & 0 < -x < 1 \\ 0, & 1 < -x < 2 \end{cases} = \begin{cases} 0, & 2 > x > 1 \\ 1-x, & 1 > x > 0 \\ 1+x, & 0 > x > -1 \\ 0, & -1 > x > -2 \end{cases} = \begin{cases} 0, & -2 < x < -1 \\ 1+x, & -1 < x < 0 \\ 1-x, & 0 < x < 1 \\ 0, & 1 < x < 2 \end{cases} = f(x)$

Hence, $f(x)$ is even function, we have $b_n = 0$

$$\begin{aligned}
\therefore f(x) &= \frac{a_0}{2} + \sum a_n \cos \frac{n\pi x}{l} \\
&= \frac{a_0}{2} + \sum a_n \cos \frac{n\pi x}{2}
\end{aligned}$$



$$a_0 = \frac{2}{l} \int_0^l f(x) dx = \int_0^2 f(x) dx = \int_0^1 (1-x) dx + \int_1^2 0 \cdot dx = \left[x - \frac{x^2}{2} \right]_0^1 = \frac{1}{2}$$

$$\begin{aligned}
a_n &= \frac{2}{l} \int_0^l f(x) \cos \frac{n\pi x}{l} dx = \frac{2}{2} \int_0^2 f(x) \cos \frac{n\pi x}{2} dx \\
&= \int_0^1 (1-x) \cos \frac{n\pi x}{2} dx + \int_1^2 0 \cdot dx \\
&= \left[(1-x) \left(\frac{2}{n\pi} \sin \frac{n\pi x}{2} \right) - (-1) \left(-\frac{4}{n^2\pi^2} \cos \frac{n\pi x}{2} \right) \right]_0^1 \\
&= \left[\left(0 - \frac{4}{n^2\pi^2} \cos \frac{n\pi}{2} \right) - \left(0 - \frac{4}{n^2\pi^2} \right) \right] \\
&= \frac{4}{n^2\pi^2} \left(1 - \cos \frac{n\pi}{2} \right)
\end{aligned}$$

$$\therefore f(x) = \frac{1}{4} + \frac{4}{\pi^2} \sum \frac{1}{n^2} \left(1 - \cos \frac{n\pi}{2} \right) \cos \frac{n\pi x}{2}$$

8. Find the Fourier expansion for $f(x) = x - x^2, -1 < x < 1$

Solution: The given function is difference between odd and even functions which can be written as

$$f(x) = f_1(x) - f_2(x)$$

where, $f_1(x) = x$, is an odd function and $f_2(x) = x^2$, is an even function

Here $l = 1$

Now, for $f_1(x) = x$ which is odd, $a_0 = 0, a_n = 0$ and

$$\begin{aligned}
b_n &= \frac{2}{l} \int_0^l f(x) \sin \frac{n\pi x}{l} dx = \frac{2}{1} \int_0^1 x \sin n\pi x dx \\
&= 2 \left[x \left(-\frac{1}{n\pi} \cos n\pi x \right) - (1) \left(-\frac{1}{n^2\pi^2} \sin n\pi x \right) \right]_0^1 \\
&= 2 \left[-\frac{1}{n\pi} (-1)^n \right] = -\frac{2}{n\pi} (-1)^n
\end{aligned}$$

Further for $f_2(x) = x^2$ which is even, $b_n = 0$

$$a_0 = \frac{2}{l} \int_0^l f(x) dx = \frac{2}{1} \int_0^1 x^2 dx = \frac{2}{3}$$

$$\begin{aligned} a_n &= \frac{2}{l} \int_0^l f(x) \cos \frac{n\pi x}{l} dx = 2 \int_0^1 x^2 \cos n\pi x dx \\ &= 2 \left[x^2 \left(\frac{1}{n\pi} \sin n\pi x \right) - 2x \left(-\frac{1}{n^2\pi^2} \cos n\pi x \right) + 2 \left(-\frac{1}{n^3\pi^3} \sin n\pi x \right) \right]_0^1 \\ &= \frac{4}{n^2\pi^2} (-1)^n \end{aligned}$$

$$\therefore f(x) = f_1(x) - f_2(x) = -f_2(x) + f_1(x)$$

$$\therefore f(x) = -\frac{1}{3} - \frac{4}{\pi^2} \sum \frac{(-1)^n}{n^2} \cos n\pi x - \frac{2}{\pi} \sum \frac{(-1)^n}{n^2} \sin n\pi x$$